

LLM/NLP 101

Large Language Models (LLMs) – Natural Language Processing (NLP)

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Why this matters

Since AI is part of our day-to-day work, it's important to be educated on it



Shared Knowledge

Common terms and how AI works behind the scenes



Better Prompts

Get more useful results from the tools we use



Risks and Limits

Know where AI falls short and why

Learning Objectives

By the end of this session, you'll be able to:

1

Define

AI technologies

2

Describe

How an LLM works

3

Recognize

Common AI limitations

4

Apply

Habits to use tools effectively

Purple Mai'a AI Learning Hub

• FOR OUR PARTNERS

LLM/NLP 101

How Purple Mai'a talks about AI — what these systems actually are, what they are not, and where a person has to stay in the loop.

🕒 ~30 min briefing 📍 Partner orientation

[Start with Module 1](#)

[See the run of show](#)

We get asked a lot of AI questions, and we would rather answer them the same way every time. This is the baseline our whole staff works from, published openly so you can read it before a meeting instead of taking our word for it in one. No math, no product pitch. Start anywhere.

Your progress



3 of 19 lessons done

Tick lessons off as you go. Saved in this browser only — nothing is uploaded.

[Reset progress](#)

READ TIME

~30 min

MODULES

3

COST

Free, no login

REUSABLE

Fork it

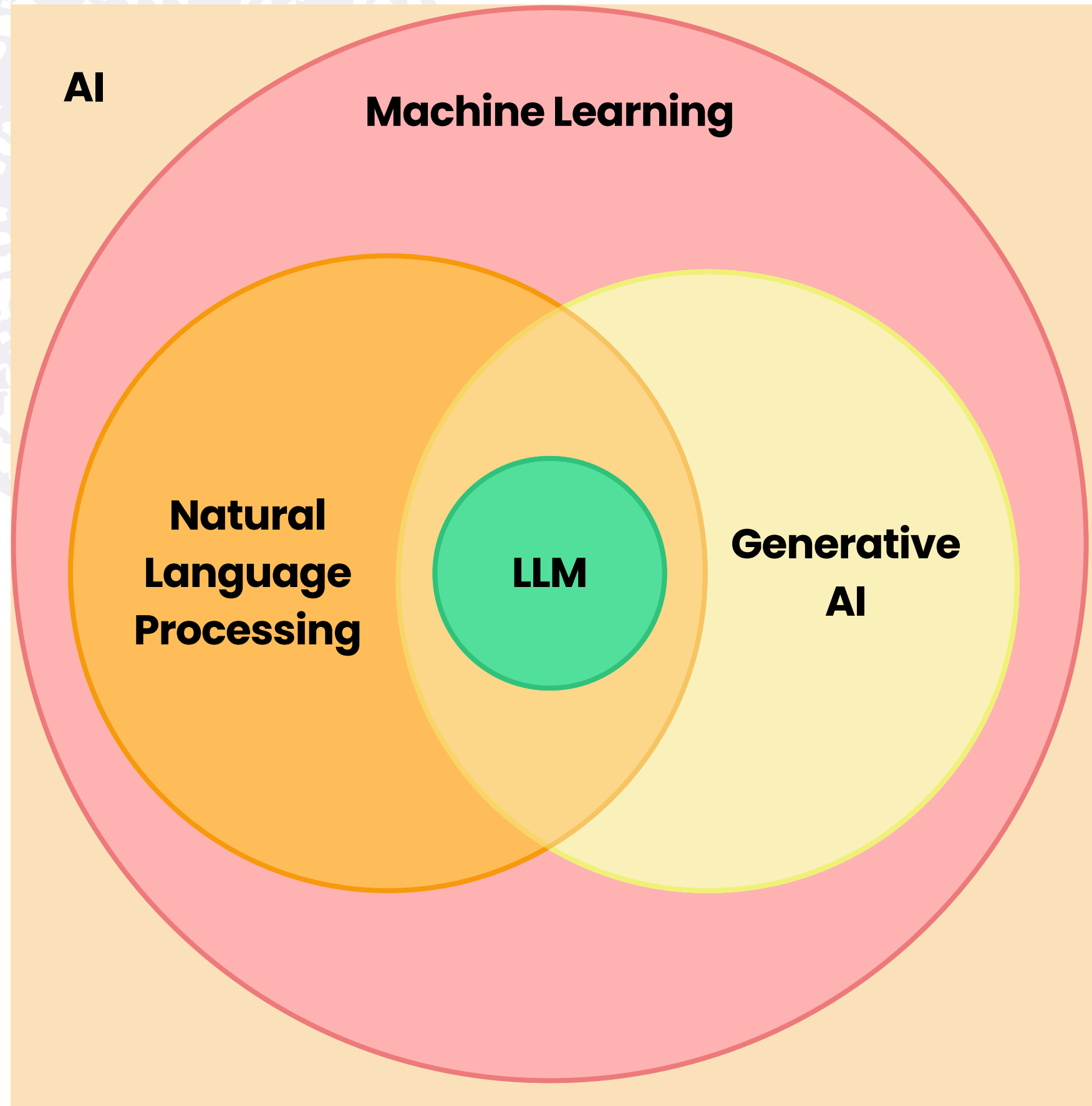


Overview:

ML, NLP, Generative AI, LLMs

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Machine Learning

A way of building AI. Systems learn patterns from data instead of being explicitly programmed rule-by-rule

Natural Language Processing (NLP)

ML applied specifically to language. Focused on getting computers to read, understand, and produce human language.

Generative AI

Creates new content (audio, text, images, video), rather than just classifying or predicting

Large Language Models (LLMs)

Specific technology behind most generative text AI today (e.g. ChatGPT, Claude, Gemini)

Large Language Models (LLMs)

Where LLM's play a role during your day



Drafting & editing emails



Summarizing long documents



Quick how-to help



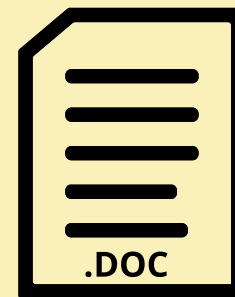
Session prep

Generative AI

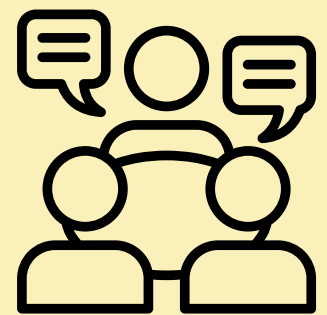
Where Generative AI plays a role during your day



Social graphics



First-draft copy



Meetings summaries

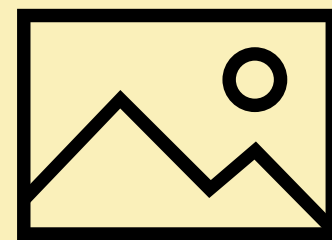
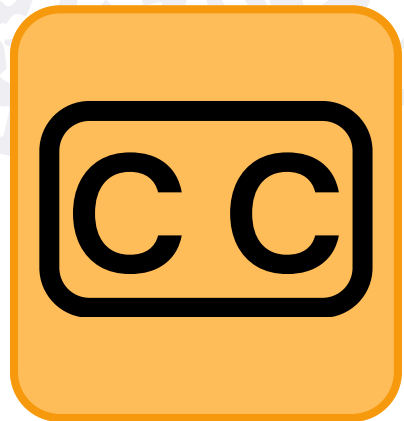


Image generation

Natural Language Processing (NLP)

Where NLP plays a role during your day



Live captions &
transcription



Translation tools

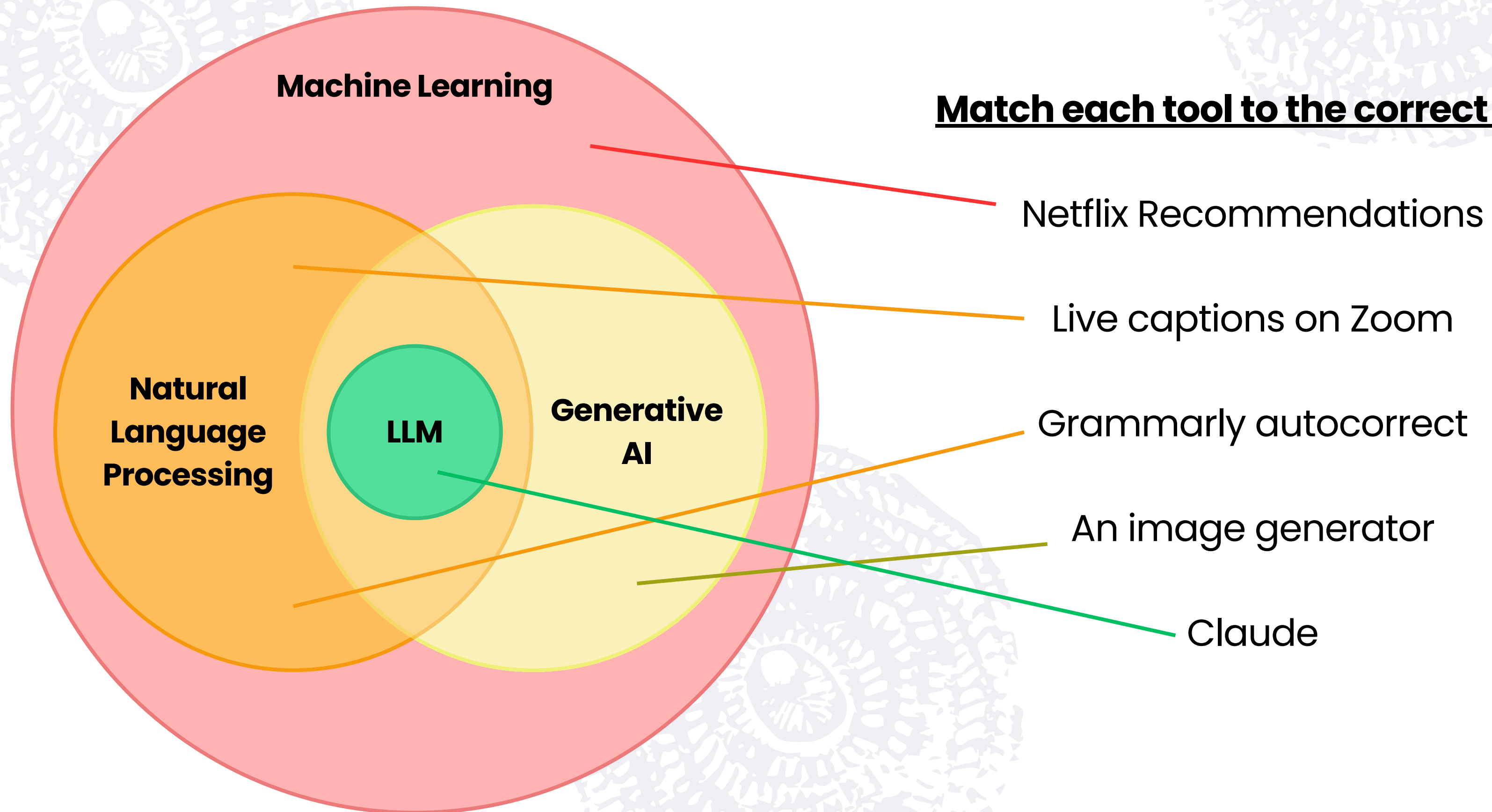


Grammar & spell-check



Voice-to-text

Check Your Understanding #1

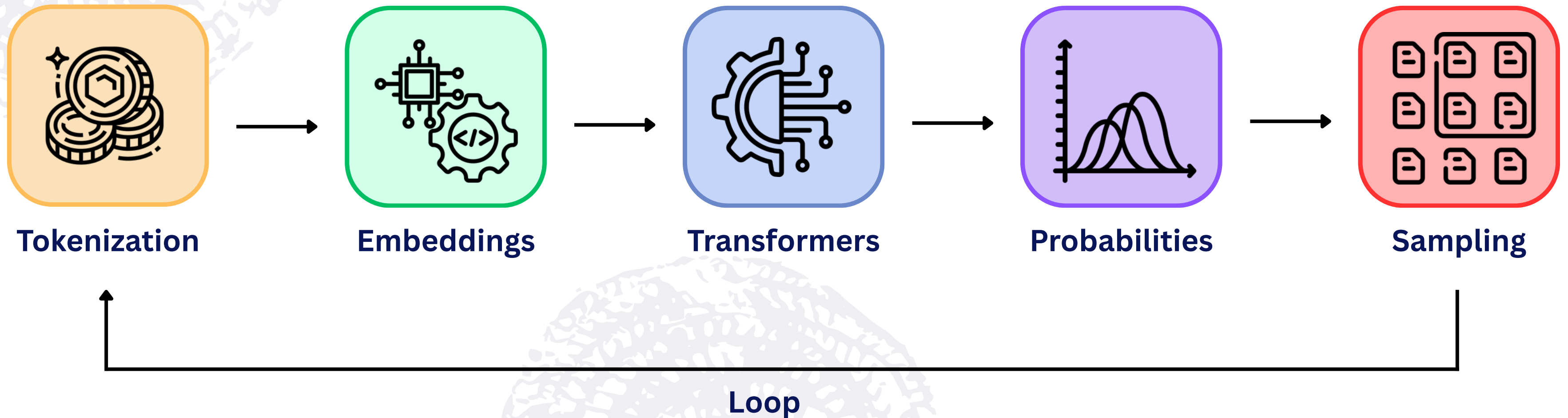


How an LLM works

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How an LLM Works

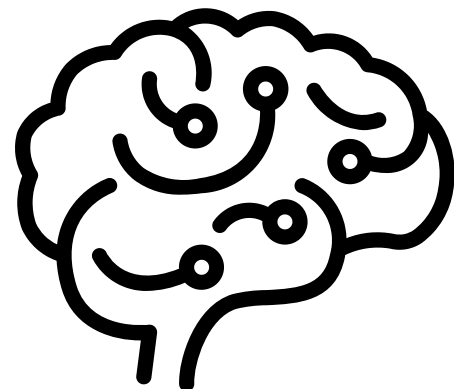


Pre-Training vs. Inference

Pre-Training

First phase of an AI model. Happens once.

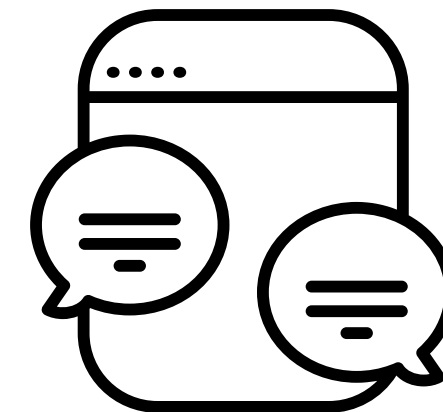
The model reads an enormous amount of text and adjusts its parameters to get better at predicting language.



Inference

Uses the model. Happens in real-time.

Every time you send a message, the model applies what it already learned.



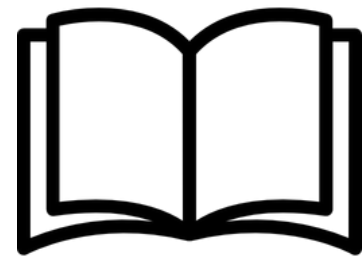
Training Data

PRE-TRAINING

The massive body of data an LLM reads.



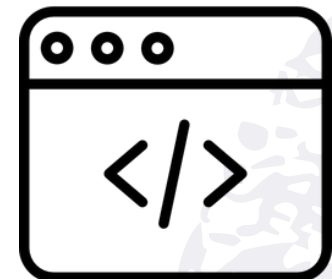
Websites



Books



Articles



Code

It's the model's source of both knowledge and gaps in information.

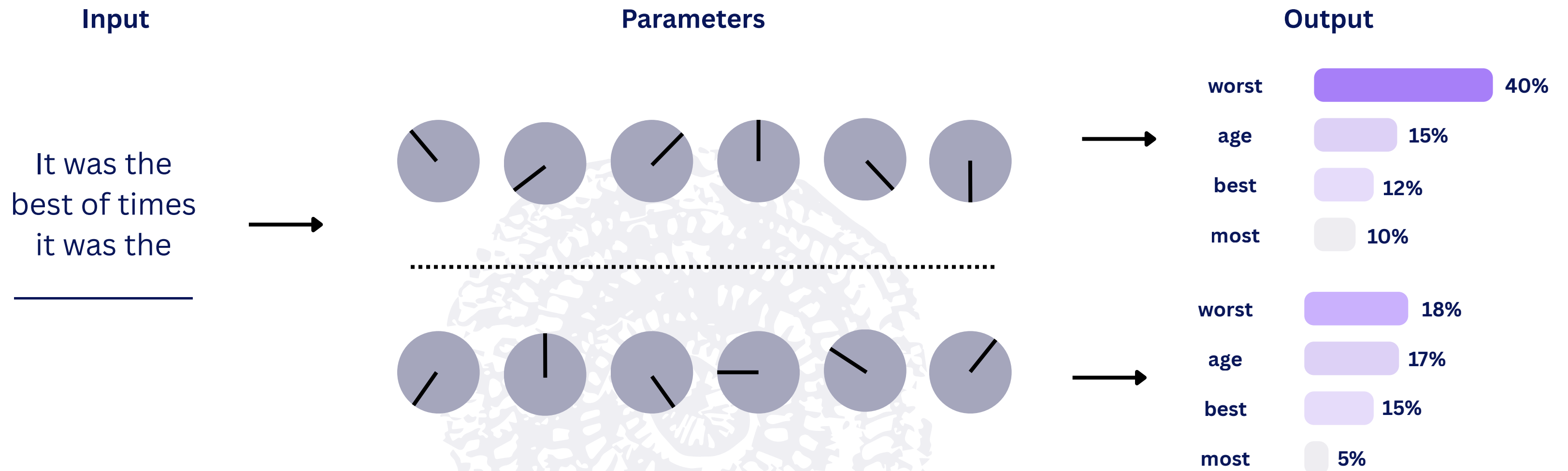
If something is underrepresented in training data (e.g., 'Ōlelo Hawai'i), the model will be weaker on it.

Parameters

PRE-TRAINING

The model's learned settings or internal "knobs" that determine how it weighs and combines information.

Changing these parameters will change the probabilities for the next word.



Check Your Understanding #2

Q: Where does an LLM's knowledge come from?

- A. It looks things up on the internet live
- B. Its training data
- C. It asks other people
- D. Its training data and user-inputted prompts

Answer: B. Its training data

Everything an AI model knows comes from the text it read during training.

Check Your Understanding #2

Q: An AI model learns from you while you chat with it.

- A. True
- B. False

Answer: B. False

Learning happens during pre-training. When you chat, that's inference, and the model applies what it already learned.

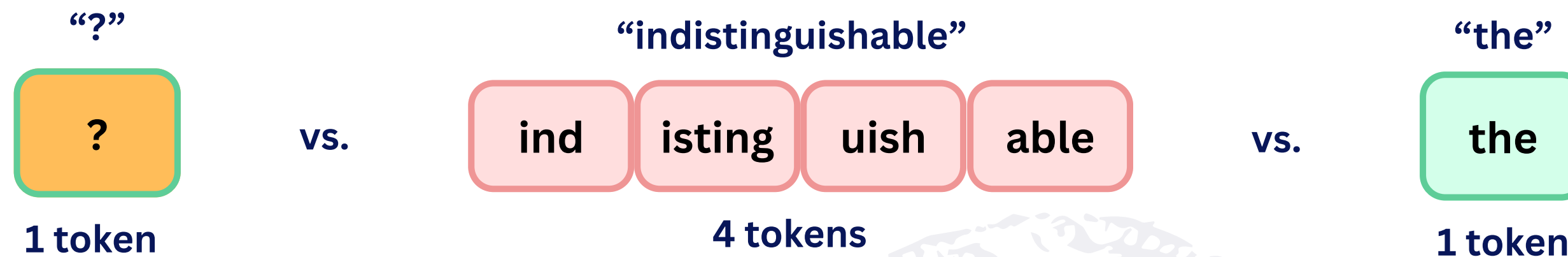
Tokenization

INFERENCE

Before a model can process anything, text is chopped into pieces called **tokens**. Every token is assigned a number (**token ID**).



Punctuation, parts of long words, and small words



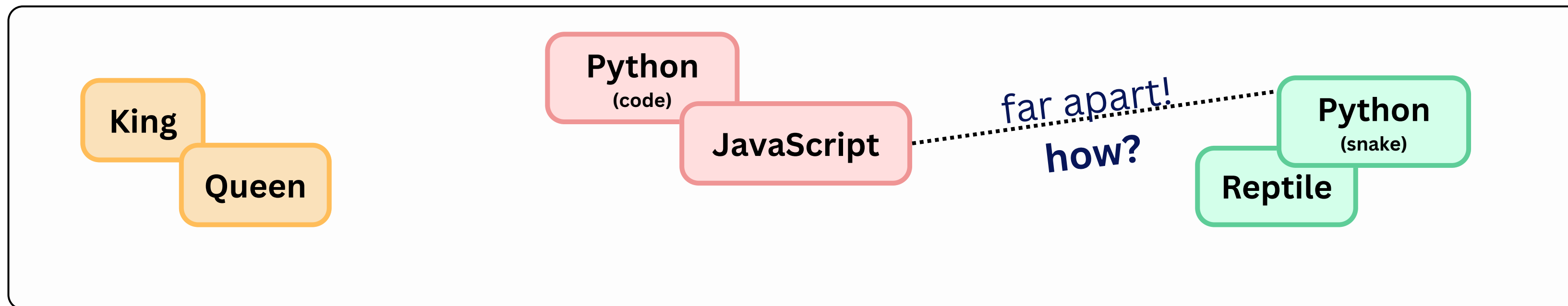
Embedding

INFERENCE

Encoder: An AI model that translates token ids to embeddings

Embedding (vector) : Sequence of numbers that represent “meaning”

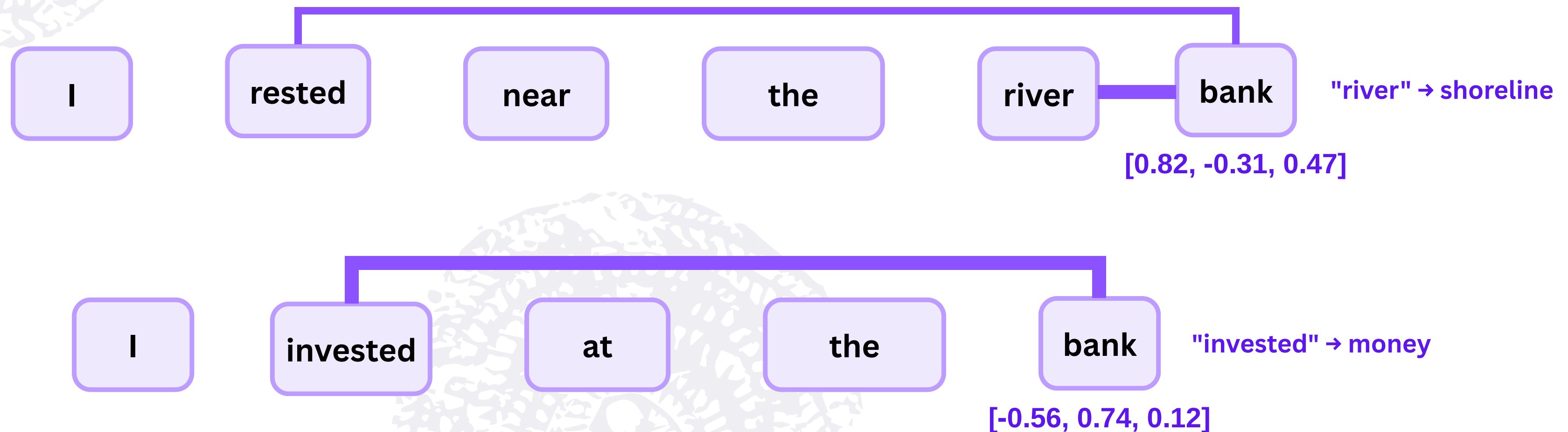
King	Queen	Python (code)	JavaScript	Python (snake)	cobra
0.81	0.78	-0.52	-0.49	0.12	0.15
0.62	0.66	0.88	0.84	-0.71	-0.68
-0.45	-0.41	0.30	0.35	0.83	0.79



Transformers

INFERENCE

For every token, the model decides which tokens matter most.
Thicker lines below mean stronger **attention**.



Next-Token Prediction

INFERENCE

Given everything so far, the model guesses the single most likely next token.

The

honu

swam

toward

the



ocean



42% ✓

shore



21%

reef



15%

current



12%

sky



10%

Probabilistic vs. Deterministic Sampling

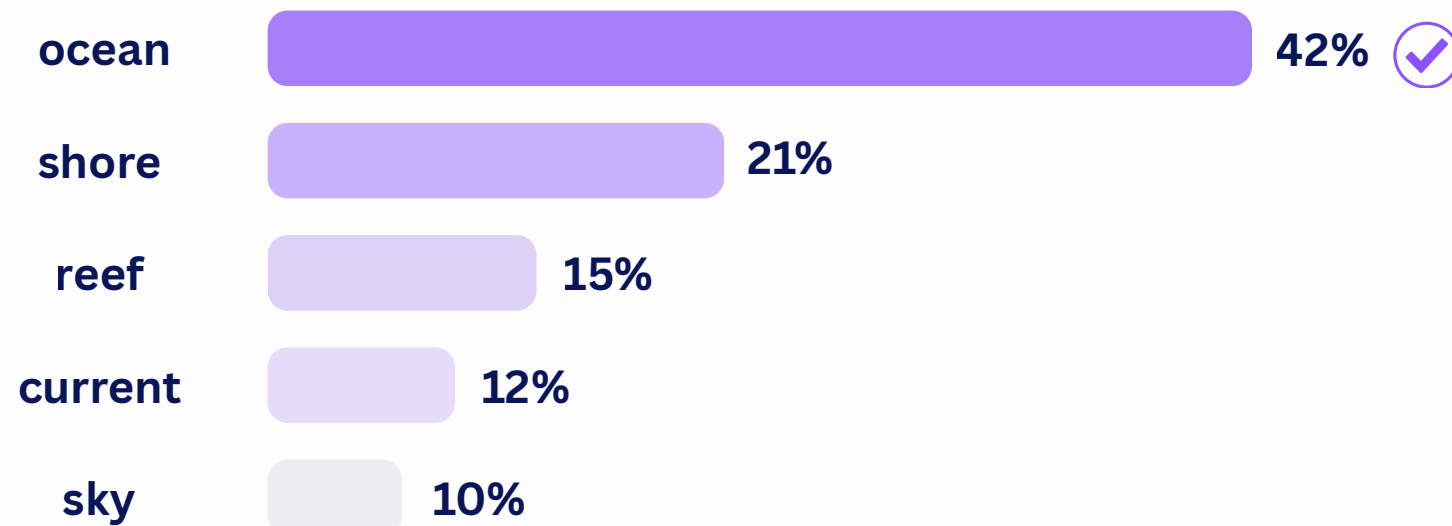
INFERENCE

A **probabilistic** system uses statistical likelihood to produce various outputs for the same input (like an LLM).

A **deterministic** system produces the same output every time for the same input (like a calculator).

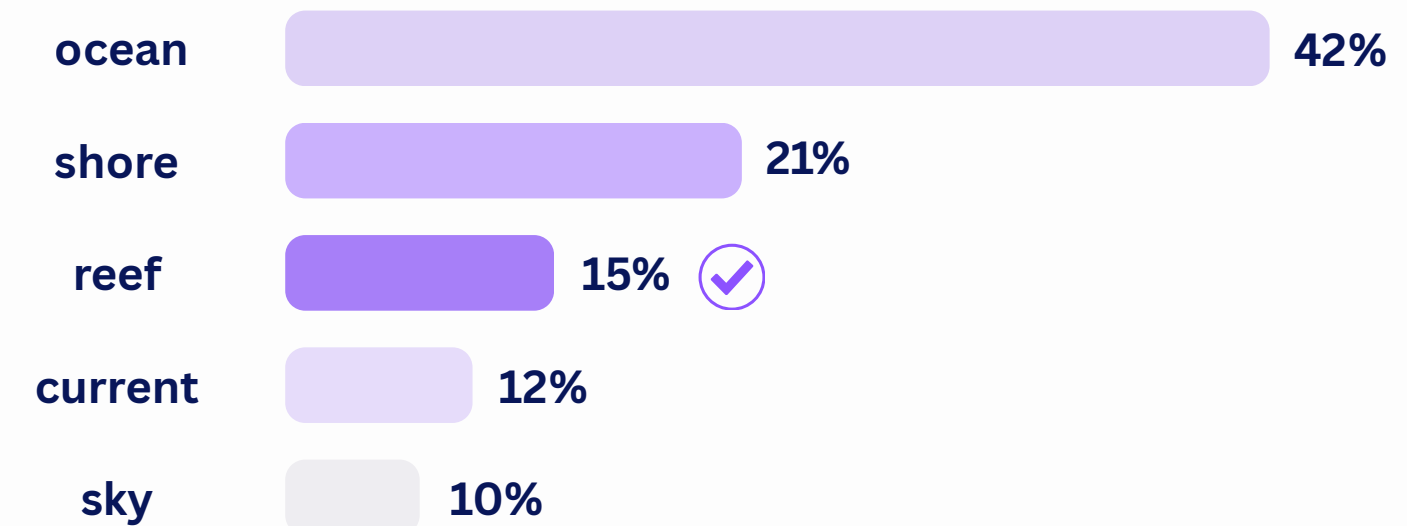
Greedy Decoding

Pick the highest probability token every time.



Sampling (Temperature 0.9)

Temperature adjusts how confident the distribution is.

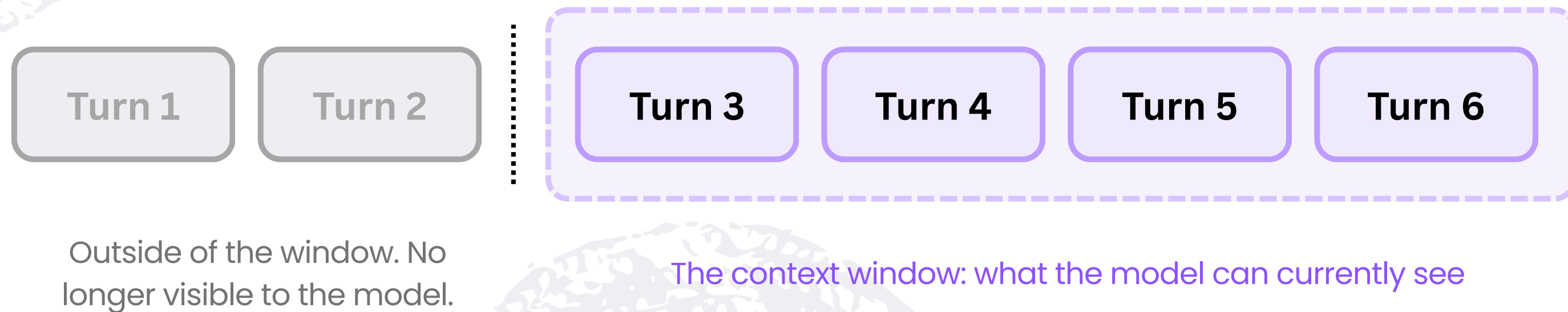


Context Windows

INFERENCE

The amount of text a model can "see" and consider at once.

This includes the prompt, your conversation so far, and any documents you've shared.



Once something falls outside the window, the model can't see it anymore.
This is why very long conversations can seem to make it "forget."

Check Your Understanding #3

Q: What is the order in which an LLM works?

- A. Tokenization → Embeddings → Transformers → Probabilities → Sampling
- B. Embeddings → Transformers → Tokenization → Probabilities → Sampling
- C. Tokenization → Transformers → Embeddings → Sampling → Probabilities
- D. Sampling → Probabilities → Embeddings → Tokenization → Transformers

Answer: A. Tokenization → Embeddings → Transformers → Probabilities → Sampling

It follows this sequence of steps then loops again for the next token.

Using LLMs intelligently

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Prompts and context

Treat the model like a capable new hire who has vast knowledge but knows nothing about the context of your prompt.

Weak Prompt

Write a bio for Keiko

The model has to guess the audience, length, tone and facts.

Strong Prompt

Write a **100-word bio** for Keiko, our program director, for a **grant funder audience**, **warm but concise**, drawing on **the notes below**.

Format

Audience

Tone

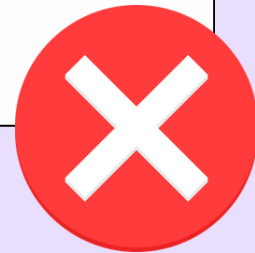
Source Material

Hallucinations

Hallucination: A misleading or incorrect response that AI presents as fact.

What page is this quote on in the 2025 report?

That quote is on **page 47**.



Why this happens

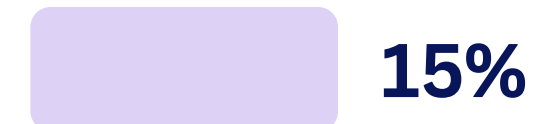
Possible next tokens after "page"

47



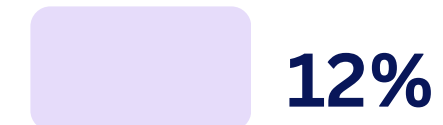
40%

100



15%

23



12%

4



10%

Grounding and verification

Grounding: Pairing a model with real, retrievable sources to reduce made-up answers.

Retrieval-Augmented Generation (RAG): A technique that lets AI search facts in an external knowledge base before writing an answer.

Plain Chat Answer

The application is due March 17.



No sources

Grounded Answer

The application is due March 17.



[GrantGuidelines.pdf, page 2](#)



Click through sources and verify information

Check Your Understanding #4

Q: Why do LLMs hallucinate?

- A. They're trying to trick you
- B. They forgot information and are trying to fill in the gaps
- C. They predict plausible words rather than verifying the truth
- D. They lack information about specific topics

Answer: C. They predict plausible words rather than verifying the truth

An LLM works by predicting the most probable next word it generates. It doesn't check whether what it's saying is true.

Bias

Models echo whose voices are online.

What goes in

**Western
Sources**

**Web
Pages**

**Published
articles**

● **‘Ōlelo Hawai‘i
Native sources**

And What comes out

Sure! Here's the plan...

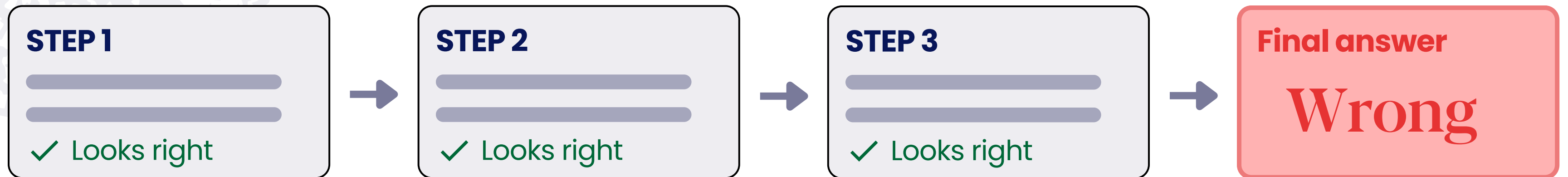


Confident, Generic, outside reasoning

Better: Defer to community
& native sources

Reasoning

Tidy steps aren't proof of a right answer.



What it is

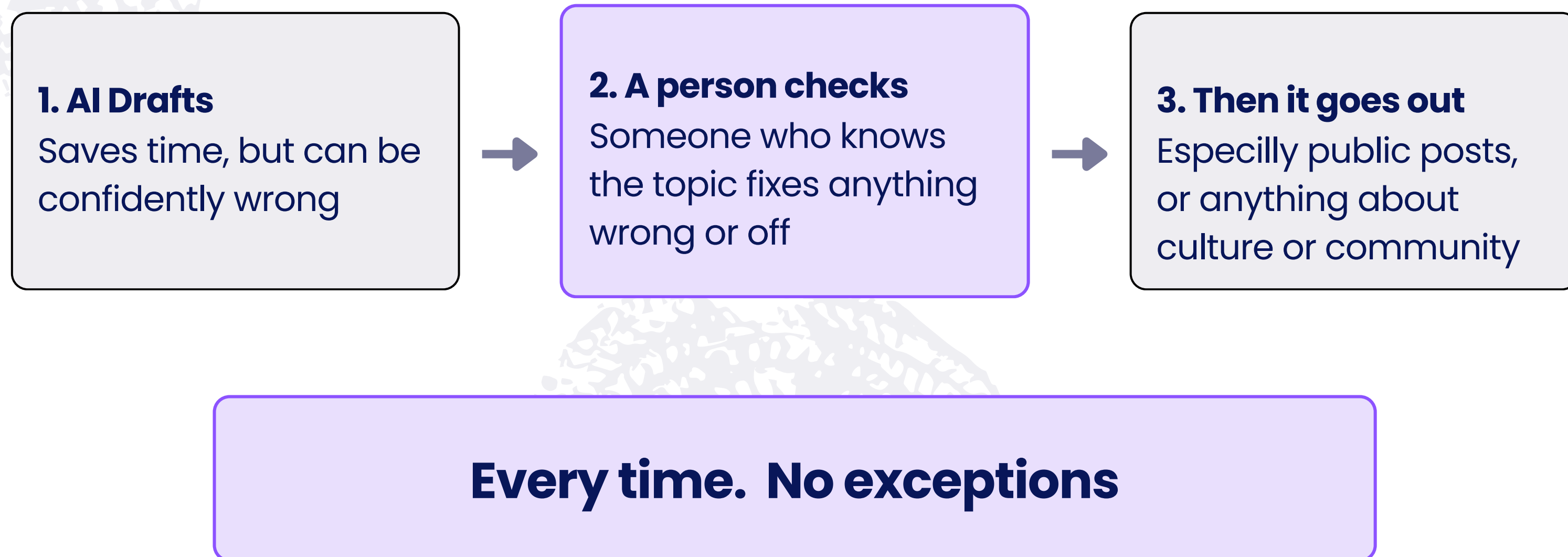
Plausable-sounding text, generated one step at a time

What it's not

Proof it's correct, or human reasoning and understanding

Human judgment and oversight

AI drafts, a person who knows the topic checks.



Proprietary models and model families

Each company builds its own line of models, released over time.

OpenAI

GPT

GPT-4



GPT-5

Anthropic

Claude

Claude Sonnet 5



Claude Sonnet 5.5

Google

Gemini

Gemini 3.7 Flash



Gemini 3.8 Flash

Same idea as **iPhone 15 → iPhone 16**: new versions from the same company

Check Your Understanding #5

Q: Where does bias in an LLM come from?

- A. The training data it learned from
- B. The user-inputted prompts
- C. The model forms its own opinions over time
- D. The parameters the model is set on

Answer: A. The training data it learned from

The model learns from huge amounts of text that already reflect whose voices, languages, and stories are well represented online, and whose aren't.

Mahalo!

Questions?

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